

Forma Studios PaaS Solution – Group 1

Presented by: Data Centre Academy (DCA)
(**Michael, Bradley, Hedieh, Saeed, Stefhanus**)

A complete cloud-based platform deployment for Forma
Studios.

Project Overview

- Environment built for Forma Studios as a PaaS solution
- 3 client workstations hosted on a Hyper-V cluster
- Redundancy via Failover Clustering
- Centralized storage with TrueNAS
- Backup and Recovery using Vinchin
- Remote access provided through SSTP VPN

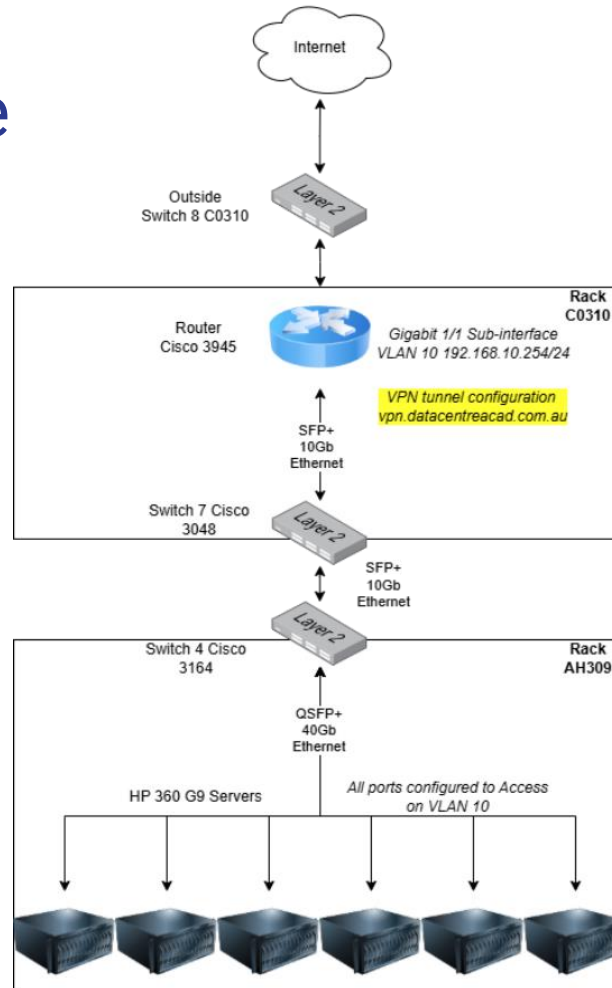
Infrastructure Summary

- VLAN 10: Production (192.168.10.0/24)
- VLAN 99: Management/iLO 92.168.99.0/24)
- Domain: FS.local

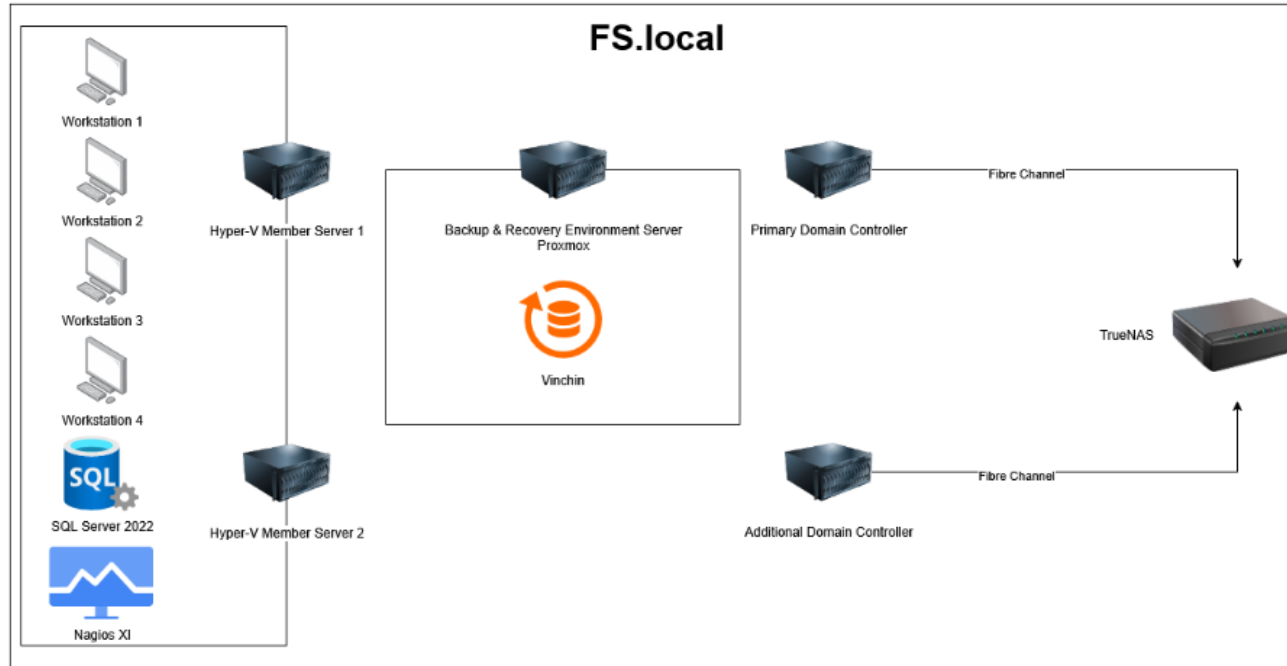
Servers:

- PDC & ADC – Domain Controllers
- VS1 & VS2 – Hyper-V Cluster Nodes
- DB – SQL Server 2022
- NAS – TrueNAS File Server
- BRE – Backup & Recovery Server (Proxmox + Vinchin)

Networking Architecture

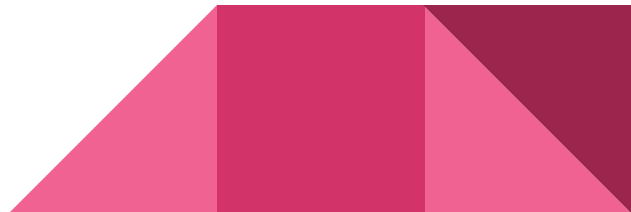


High Level System Architecture



Virtual Machines

- Workstations (VM1, VM2, VM3, VM4)
- Nagios (VM5, MON)
- SQL Server (DB)
- Vinchin (VOS)



Storage Setup (TrueNAS)

TrueNAS configured with RAID 10 (4 drives)

3 LUNs:

- 1 GB Disk Witness (Cluster quorum)
- 1.6 TB Shared Storage (Accounts, Web, Corporate)
- 2 TB Backup Folder

Hyper-V Cluster (VS1 & VS2) connects via iSCSI initiator
All client data stored centrally on \\RFS\Corporate

Hyper-V NAS (HVNAS)

- 1 iSCSI connection (Clustered Share)
- 2 SMB Shares (VM_Apps & Folder Witness)

\\hvnas\VM_Apps

\\hvnas\\$\ClusteredShare\Volume1\

Has a little SD card (128GB)
Two 1.8 TB SAS in a mirror

Domain Roles

PDC	ADC
DNS	DNS
DHCP	DHCP
Failover Clustering	Failover Clustering
Active Directory Domain Services	Active Directory Domain Services
Network Policy Server	
Direct Access and VPN (RAS)	

Failover and Clustering

- VS1 and VS2 form a Hyper-V failover cluster
 - Shared storage on TrueNAS allows live migration
 - If VS1 fails, VMs auto-start on VS2
 - Managed through Failover Cluster Manager on PDC or ADC
 - Testing: VM migration between nodes confirmed high availability
- 

Domain Group Policies

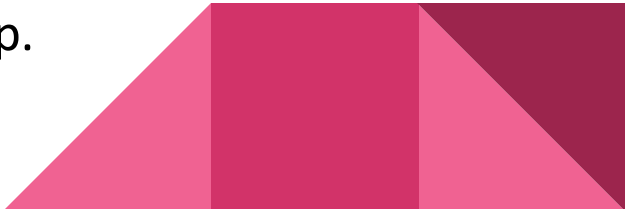
Users:

- John – Accountant (Access to Accounts folder)
- Jack – Web Developer (Access to Web folder + SQL)
- Jill – General User (Access to Corporate folder)

Mapped Drives via GPO:

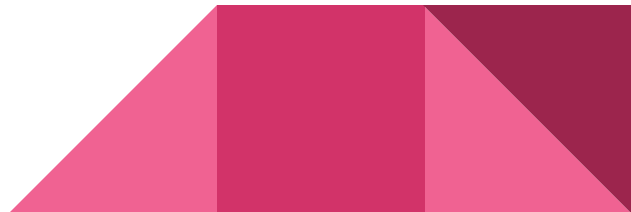
- A: → \\rfs\Corporate\Accounts (Accountant)
- W: → \\rfs\Corporate\web (Web Developer)
- H: → \\rfs\Corporate (All users)

Applied using Item-Level Targeting for each group.



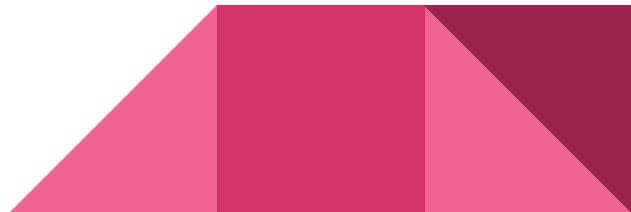
GP: Allow Pings (ICMPv4)

- Create firewall rule for all domain computers to allow ICMPv4 from local IP addresses AND remote IP addresses in the scope:
 - 192.168.99.1 - 192.168.99.254



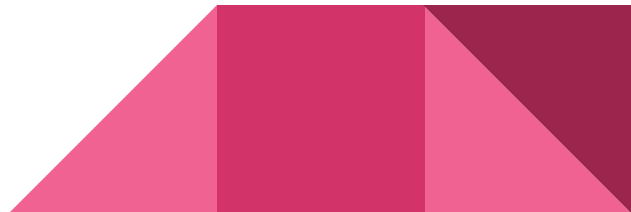
GP: Allow more than 1 RDP Session

- Limit number of connections
 - Set to 5 (Doesn't matter unless you have a CAL)
- Restrict Remote Desktop Services users to a single Remote Desktop
 - Set to Disabled



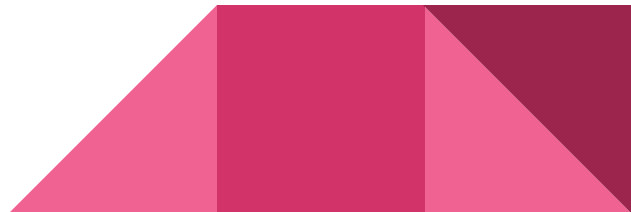
GP: Link DNS servers to all domain computers

- Under Computer Configuration -> Policies -> Administrative Templates -> Network -> DNS Clients -> DNS Servers
- Enable, and add “192.168.10.1, 192.168.10.2” to the IP Addresses.



GP: Allow Local PowerShell Scripts

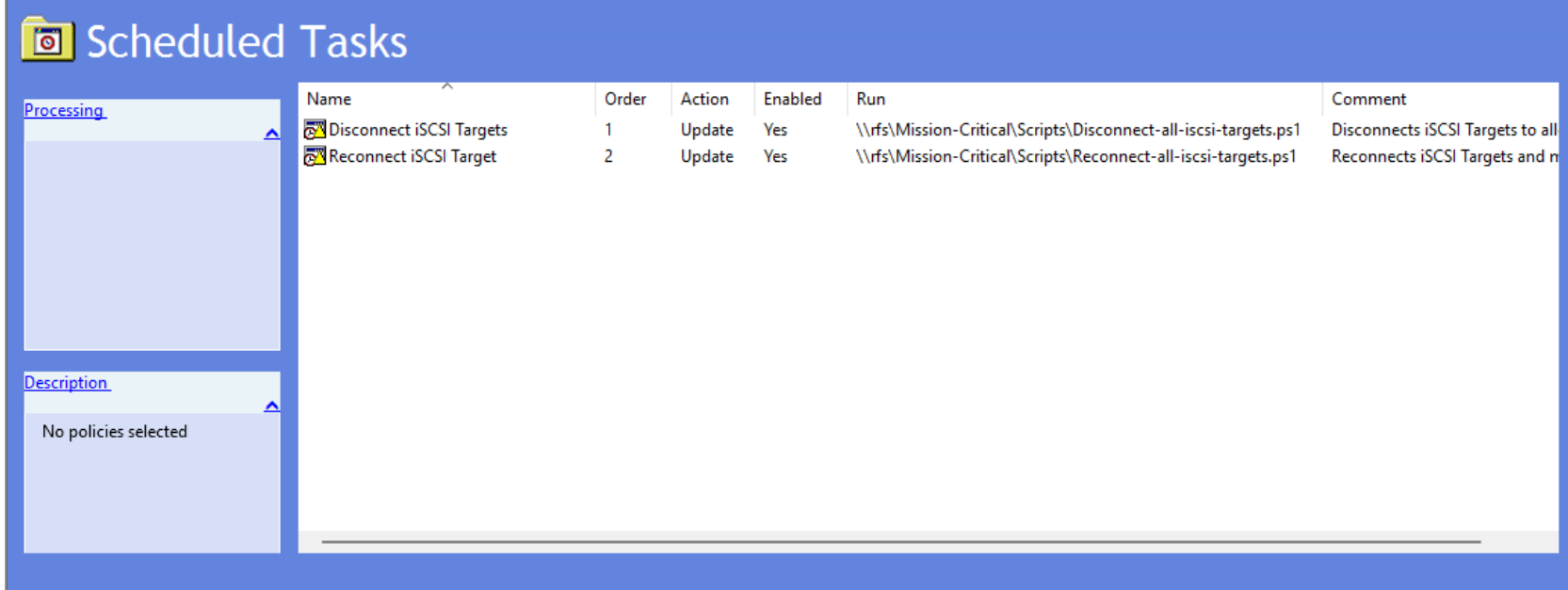
- Under Computer Configuration -> Policies -> Administrative Templates -> Windows Components -> Windows PowerShell
- Set the following settings:
 - Turn on PowerShell Script Block Logging -> Disabled
 - Turn on PowerShell Execution -> Enabled
 - Set “Execution Policy” to “Allow local scripts and remote signed scripts”



GP: Allow Local PowerShell Scripts Pt2

- Create self-signed certificate:
 - `New-SelfSignedCertificate -DnsName "fs.local" -CertStoreLocation Cert:\LocalMachine\My -Subject "Local Powershell Scripts" -KeyAlgorithm RSA -KeyLength 2048 -Provider "Microsoft Enhanced RSA and AES Cryptographic Provider" -KeyExportPolicy Exportable -KeyUsage DigitalSignature -Type CodeSigning`
- You can see the certificate within Cert manager under Local Machine -> Personal.
- **Copy the certificate to root trusted folder.**
- Save the long ID as an environment variable called "SelfSignCert", expires within a year. You can reference this in PowerShell by calling `$env:SelfSignCert`
- Sign scripts by running:
 - `Set-AuthenticodeSignature -FilePath .\{SCRIPT HERE}.ps1 -Certificate (Get-ChildItem Cert:\LocalMachine\My\$env:SelfSignCert)`

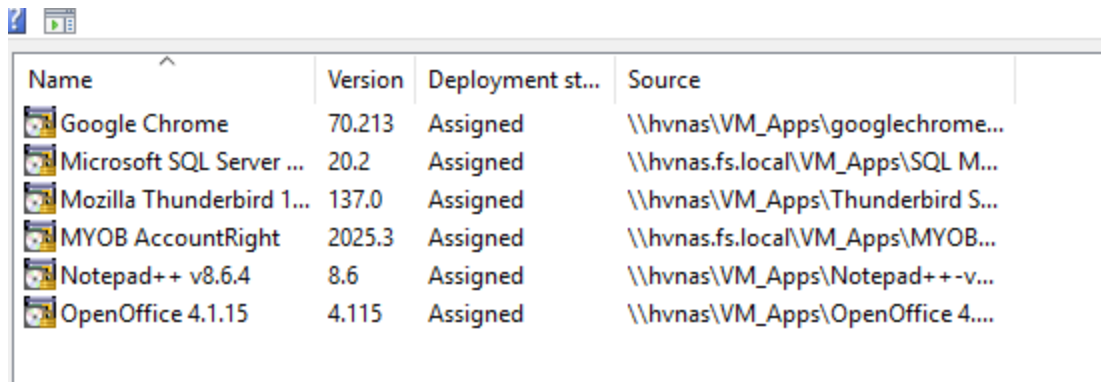
GP: Turn on/off iSCSI for Vinchin



The screenshot displays the Windows 'Scheduled Tasks' window. The left-hand pane contains two sections: 'Processing' and 'Description'. The 'Processing' section is currently selected and shows a list of tasks. The 'Description' section below it indicates 'No policies selected'. The main pane on the right displays a table of scheduled tasks.

Name	Order	Action	Enabled	Run	Comment
 Disconnect iSCSI Targets	1	Update	Yes	\\rfs\Mission-Critical\Scripts\Disconnect-all-iscsi-targets.ps1	Disconnects iSCSI Targets to all
 Reconnect iSCSI Target	2	Update	Yes	\\rfs\Mission-Critical\Scripts\Reconnect-all-iscsi-targets.ps1	Reconnects iSCSI Targets and m

GP: Application Install Policy

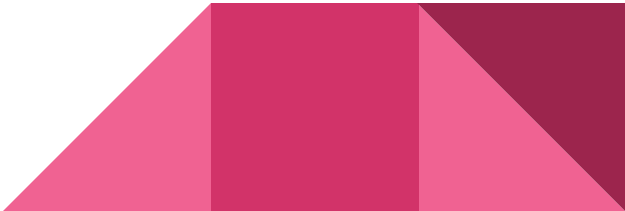


A screenshot of a Windows File Explorer window displaying a list of applications. The window has a standard Windows interface with a title bar and a toolbar. The list is organized into columns: Name, Version, Deployment status, and Source. Each row includes a small application icon on the left. The applications listed are Google Chrome, Microsoft SQL Server, Mozilla Thunderbird, MYOB AccountRight, Notepad++, and OpenOffice. All applications are marked as 'Assigned' in the deployment status column.

Name	Version	Deployment st...	Source
 Google Chrome	70.213	Assigned	\\hvnas\VM_Apps\googlechrome...
 Microsoft SQL Server ...	20.2	Assigned	\\hvnas.fs.local\VM_Apps\SQL M...
 Mozilla Thunderbird 1...	137.0	Assigned	\\hvnas\VM_Apps\Thunderbird S...
 MYOB AccountRight	2025.3	Assigned	\\hvnas.fs.local\VM_Apps\MYOB...
 Notepad++ v8.6.4	8.6	Assigned	\\hvnas\VM_Apps\Notepad++-v...
 OpenOffice 4.1.15	4.115	Assigned	\\hvnas\VM_Apps\OpenOffice 4....

GP: Increase MaxEnvelopeSizekb

New Registry Key in Computer Config -> Preferences -> Windows Settings -> Registry

- Set:
 - Action: Update
 - Hive: HKEY_LOCAL_MACHINE
 - Key Path: SOFTWARE\Microsoft\Windows\CurrentVersion\WSMan\Client
 - Value name: maxEnvelopeSize
 - Notice it excludes "kb" at the end
 - Value type: REG_DWORD
 - Value data: 8192
 - Base: Decimal
- 

Backup and Disaster Recovery

Backup handled by Vinchin on BRE server

Backup Schedule:

- Daily Incremental (varies between 9pm - 6am)
- Full Back up every week

Stored on 1.8TB WarmSite volume

Restore testing performed successfully using Vinchin restore feature

Recovery Time Objective (RTO): 1 Day

Backup Scheme

Sat	Sun	Mon	Tue	Wed	Thu	Fri
Incremental	Incremental	Incremental	Incremental	Incremental	Incremental	Full Backup

- Vinchin Agents are installed in PDC, ADC, VS1, VS2
- Vinchin Hyper-V Agent installed in VS1, VS2
- You guessed it, we're using Vinchin
 - We also used:
 - Veeam (Removed last second)
 - Windows Server Backup

Vinchin Settings

- All jobs have Data Compression: ON

Storage

Onsite iSCSI ([\\backups\Server-Backups\Vinchin](#))

Offsite1 ([\\PDC\Offsite1\Vinchin](#))

Offsite2 ([\\ADC\Offsite2\Vinchin](#))

Jobs

PDC to Onsite

ADC to Onsite

All VMs to Onsite

VS1 & VS2 to Onsite

NAS-Share to Onsite

NAS-Share to Offsite1

NAS-Share to Offsite2

SSTP VPN Setup

- SSTP Server installed on PDC
- SSL/TLS Certificate issued by instructor CA
- External FQDN: vpn.datacentreacad.com.au
- Clients connect using Windows VPN client (SSTP)
- After VPN connection, clients RDP to their VMs (vm1, vm2, vm3, vm4)



Scripts: Disconnect all iSCSI Targets

```
# Stop script on error
$PSNativeCommandUseErrorActionPreference = $true
$ErrorActionPreference = 'Stop'

# verify cluster status before anything frisky happens
$nodes = Get-ClusterNode -Cluster "DDC"
if ($nodes.Count -le 1) {
    #Exit since everything is broken
    Write-Error "There are less than 2 nodes available, terminating..."
    exit 1
}
```

```
# if failover clustering is using this node,
# migrate to another node.
$thisnode = hostname
$cnodes = Get-ClusterGroup -Cluster "DDC"
foreach ($cnode in $cnodes) {
    # if this machine is the active node,
    # move it to another node
    if ($cnode.OwnerNode -eq $thisnode){
        Move-ClusterGroup -Name $cnode.Name
    }
}
```

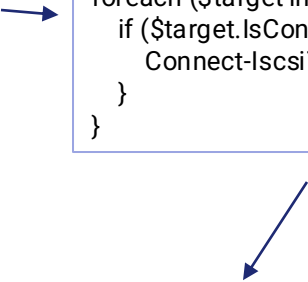
```
# disconnect iscsi initiator targets
$targets = Get-IscsiTarget
foreach ($target in $targets) {
    if ($target.IsConnected -ne $true){
        Disconnect-IscsiTarget -NodeAddress $target.NodeAddress -Confirm:$false
    }
}
```



Scripts: Reconnect all iSCSI Targets

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# Stop script on error
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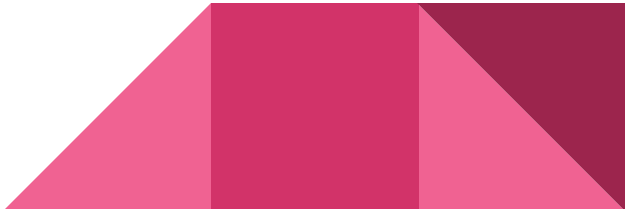
# verify cluster status before anything frisky happens
$nodes = Get-ClusterNode -Cluster "DDC"
if ($nodes.Count -le 1) {
    #Exit since everything is broken
    Write-Error "There are less than 2 nodes available, terminating..."
    exit 1
}
```



```
# connect iscsi initiator targets
$targets = Get-IscsiTarget
foreach ($target in $targets) {
    if ($target.IsConnected -ne $true){
        Connect-IscsiTarget -NodeAddress $target.NodeAddress
    }
}
```

```
# migrate to preferred active node. (if it exists)
$preferred = "PDC"
$nodes = Get-ClusterGroup -Cluster "DDC"
foreach ($node in $nodes) {
    # if this machine is the active node, move it to another node
    if ($node.OwnerNode -ne $preferred) {
        Move-ClusterGroup -Name $node.Name -Node $preferred
    }
}
```

Testing Results

- Domain logins tested for John, Jack, and Jill
 - Drive mappings confirmed for each group
 - Live migration between VS1 and VS2 successful
 - Vinchin backup and restore test completed
 - SSTP VPN access verified from external network
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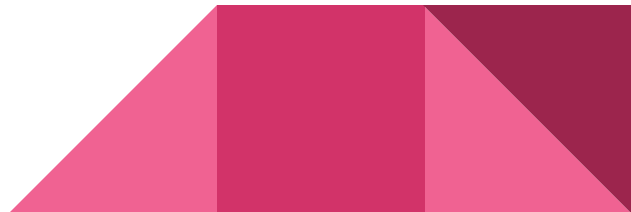
Team Reflection

- Learned how to deploy and manage Hyper-V clusters
- Configured domain roles, policies, and iSCSI storage
- Practiced data protection with Vinchin and TrueNAS
- Understood high availability and failover concepts
- Improved troubleshooting and collaboration skills



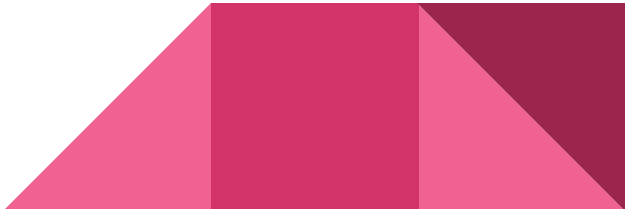
Lessons Learned:

- Backup Software tend to have issues.
- Leave Authenticated Users in Group Policy Objects
 - Required for objects to read the policy itself and apply when necessary.
- Leave Authenticated Users in Sharing & Security at the ROOT level
 - All users inherit from this role



Conclusion

Forma Studios now has a fully functional and redundant PaaS environment:

- Secure access via VPN
 - Reliable data protection
 - Centralized file storage
 - High availability and failover-ready infrastructure
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Questions?

A graphic featuring the words "thank you!" in a playful, multi-colored font. The letters are arranged in two lines: "thank" on top and "you!" on the bottom. The colors of the letters include yellow, green, blue, red, and orange. The graphic is set against a white background, which is itself a square inset on a larger blue background with a geometric pattern of triangles.